

4. EXCAVATION, LOADING AND TRANSPORT

Excavation and loading relates to excavating the bench material and loading them onto the transporting equipment. However, in some cases the excavator itself transports and dumps the bucket load (e.g. dragline, scraper). This operation may be accomplished in cyclic/discontinuous mode or in continuous mode.

Like excavation and loading, transportation of the loaded material may also be accomplished either in discontinuous/cyclic mode or in continuous mode. In this operation, the transporting equipment transport the loaded material to the respective destinations – coal/ore to coal depot/ore stock pile, and overburden/waste to overburden/waste/spoil dumps.

4.1. Discontinuous/Cyclic Methods of Excavation, Loading and Transport

The excavators and/or loaders that work in cyclic mode are all single bucket machines. These machines excavate the material directly from the face/bench (if the material is soft enough) or from the muck pile (loosened/blasted face) and load onto the transport equipment in a cyclical manner. The excavators and loaders are mainly – electric rope shovels, hydraulic shovels and backhoes, dragline and scrapers. Rope shovels and hydraulic excavators generally load the material onto dumpers for their onward transportation, whereas the dragline and scrapers transports their own load.

The dozers and front-end loaders are generally used as utility equipment, but under certain conditions dozer may also be used as primary excavator and front-end loader as primary excavator and/or loader. The cyclic mode of excavation and loading is technically applicable for all type of material.

For cyclic transport, dumpers are mainly used. However, the material may also be transported through self transport of scraper or by direct casting of dragline.

4.1.1. Shovel–Dumper Combination

4.1.1.1. Shovels

Development of Shovels

The shovel is one of the oldest types of excavating equipment. With time, the machine grew in capacity, steam power was replaced by diesel fuel and finally by electricity in the larger units. The basic configuration has changed relatively less but the efficiency and sophistication have increased substantially (Martin et.al., 1982).

The steam shovel was invented and patented by William Otis in 1839. The first machines were known as 'partial-swing', since the dipper arm could not rotate through 360 degrees. They were built on a railway chassis, on which the boiler and movement engines were mounted. The shovel arm and driving engines were mounted at one end of the chassis, which accounts for the limited swing. Temporary rail tracks were laid by workers where the shovel was expected to work, and repositioned as required. Later machines were supplied with caterpillar tracks, obviating the need for rails. The full-swing, revolving shovel was developed in England in 1884, and became the preferred format for these machines.

Perhaps the most famous application of steam shovels is the digging of the Panama Canal. One hundred and two shovels worked in that decade-long dig. Of these, seventy-seven were built by Bucyrus and the remaining ones were Marion shovels.

Steam shovels assisted mining operations - the iron mines of Minnesota, the copper mines of Chile and Montana, placer mines of the Klondike - all had earth-moving equipment. But it was with the burgeoning open-pit mines - first in Bingham Canyon, Utah – the shovels became popular, and were being used around the world from Australia to Russia to coal mines in China.

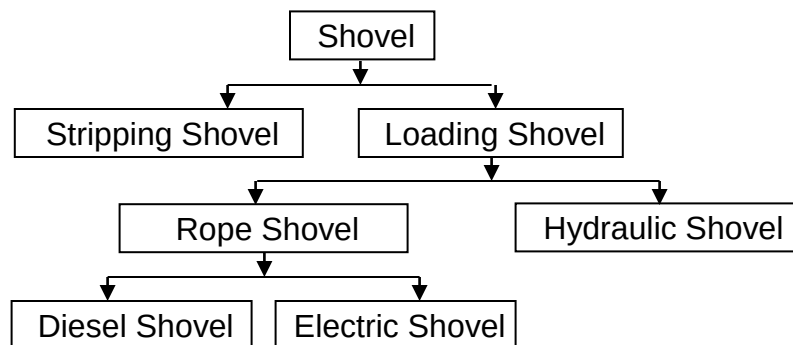
Shovels were also used for construction, road and quarry work. Steam shovels became more and more popular in the 1920s with the publicly funded road building programs around North America. Thousands of miles of State Highways were built in this time period, together with new factories, docks, ports, buildings, and grain elevators.

During the 1930s steam shovels lost out to the simpler, cheaper diesel-powered excavating shovels. Open-pit mines were electrified at this time that helped the development of electric rope shovels.

In the 1950s and 1960s Marion Shovel built massive stripping shovels for coal operations in the Eastern US. Shovels of note were the Marion 360, the Marion 5900, and the Marion 6360 with a 140 m³ bucket capacity. Bucyrus constructed one of the most famous monsters - the Big Brutus, the largest power shovel ever built and the largest still in existence. The GEM of Egypt (GEM standing for "Giant Excavating Machine" and Egypt referring to the Egypt Valley in Belmont County, eastern Ohio where it was first put to use), which operated from 1967 to 1988 and then dismantled, was of comparable size (http://en.wikipedia.org/wiki/Steam_shovel).

So far as the development of hydraulic excavators are concerned, world's first diesel powered all hydraulic crawler excavator (model B 504) was developed and presented by DEMAG at the Hannover industrial fair in the year 1954 (<http://www.komatsu-mining.de/en/content/history-komatsu-mining-germany-gmbh>).

Classification of Shovels



The stripping shovels are presently outdated. They were large size shovels with higher bucket capacities and were used for direct casting of overburden/ inter-burden to spoil heap. A classical example of stripping shovel is “Silver Spade” that worked in Ohio strip mine in Cardiz, Ohio, USA. It was built in 1965 and worked for 42 years till a major break down occurred after which it was dismantled. It was 67m (220 ft) tall with 61m (200 ft) long boom and weighed 6419 te (7000 t) . The bucket capacity was 156.7 m³ (205 yd³) and the maximum casting distance was 119m (390 ft). During its working life of 42 years, it had moved 464.1×10⁶ m³ (607×10⁶ yd³) of material.

Marion 6360, also known as The Captain, was another large stripping shovel with a bucket capacity of 140 m³ and boom length of 66m. It was used in operation by Southwestern Illinois Coal Corporation and then by Arch Coal, before it was scrapped in early 1990.

The diesel operated rope shovels were first developed in 1930s that marked the end of era of steam shovels. These shovels were more compact, efficient and independent (unlike the electric rope shovels that require a trailing cable for power supply). However, with the introduction of hydraulic shovels with diesel prime-mover the diesel rope shovels were gradually outdated and sometimes in 1990s became totally obsolete.

The shovels that are in use in the present-day surface mines are mainly of two types – electric rope shovel and hydraulic shovel. A shovel stands on the floor of the bench it excavates and generally loads onto a dumper, and they have very limited capability to dig below their crawler level. For digging below the crawler level and loading onto dumpers hydraulic backhoes are used.

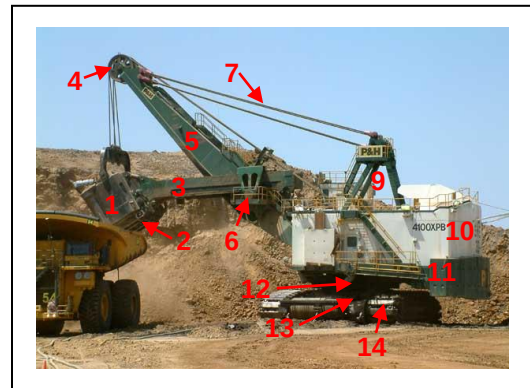
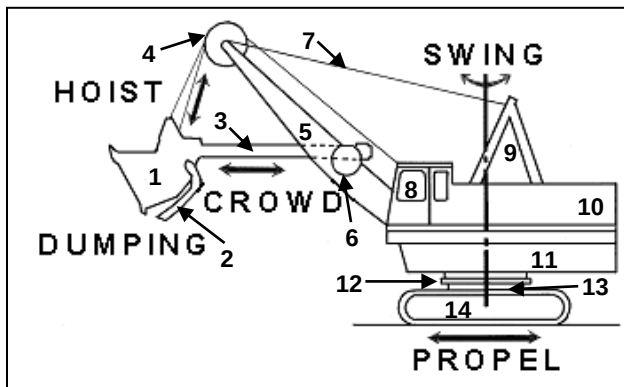
Currently P&H 4100XPC (35.9 - 76.5 m³) and Cat-7495 (30.6 - 62.6 m³) are the largest electric rope shovels built by P&H Mining Equipment and Caterpillar respectively. Cat-6120B H FS (46 - 65 m³) is the largest amongst the available hydraulic shovels. Hydraulic shovel RH 400 has a bucket capacity of 45 m³ and a maximum cutting height of 20.2 m. Cat-6060 is the largest capacity backhoe presently available with a bucket capacity of 34.0 m³ (<https://mining.cat.com/products/surface-mining/electric-rope-shovels>; <https://mining.cat.com/products/surface-mining/hydraulic-shovels>; <http://www.phmining.com/en/PHMining/Mining-Equipment/Electric-Shovels.htm>).

In Indian surface mining industry, 10 – 20 m³ electric shovels and 5 – 10 m³ hydraulic shovels are becoming more common. However, the largest size of electric shovel and the hydraulic shovel being currently used is of 42 m³ capacity (Gevra OCP, SECL, CIL) and 36 m³ capacity (Rampur-Agucha Mine of HZL, Vedanta) respectively.

General Features and Characteristics of Shovel

Electric Shovel

The electric shovels are rugged, high production units with all electric drives and a trailing cable to bring power to the machine.



POWERED FUNCTIONS → **Swing and Propel** – Motors **Hoist** – Cable/Drum **Dumping** – Gravity
Crowd – Cable/Drum or Gear **Dipper Trip** – Small Motor

1 – Bucket/Dipper	2 – Bucket Door	3 – Dipper Stick	4 – Boom Point Sheave
5 – Boom	6 – Shipper Shaft	7 – Suspension Cable	8 – Operator's Cab
9 – A Frame	10 – Machinery House	11 – Revolving Frame	12 – Swing Circle
13 – Lower Works	14 – Crawler		

General Features of Electric Shovel

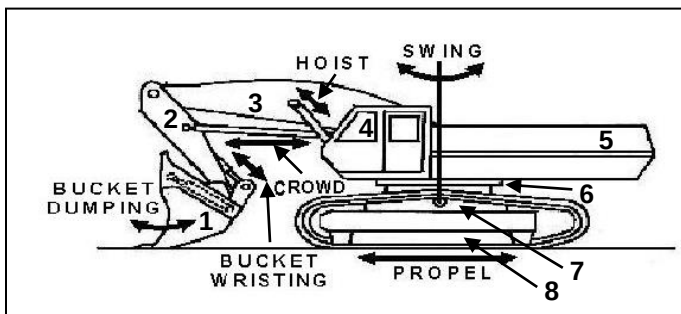
❖ Crawler mounted (two tracks)	❖ Cable or rack & pinion crowd system
❖ Fully revolving upper works	❖ Fully enclosed machinery house
❖ Cable hoist system	

General Characteristics of Electric Shovel

❖ High sustained production capability	❖ Maintenance is performed in pit
❖ High reliability and a proven design	❖ Can work in close quarters
❖ Heavy, rugged construction, suited to tough digging conditions	⇒ High capital cost and low operating cost
❖ High tooth forces on dipper, limited by machine stability	➤ Limited capability to dig below its floor level
❖ Limited gradeability and good stability	➤ Dumping action can be hard on trucks
❖ Medium ground pressures (2 to 3.5 kg/cm ²)	➤ Relatively fixed digging path
❖ Low operator fatigue	➤ Limited mobility between faces
❖ Safe operator location and good operator visibility	➤ Support equipment required to maintain working area
❖ Good blending vertically at face	➤ Low propel speeds
❖ Good performance even with bad floor conditions	➤ Limited capability for selective loading
❖ Long service life of over 20 years	➤ Moderate/high operator skill and training is required
❖ Minimal preparation-(blasting) required	➤ Altitude, temperature or humidity may reduce electrical efficiency

Hydraulic Shovel

In hydraulic shovels the operating functions are hydraulic powered. The prime-mover for these shovels is generally diesel power. However, in very high capacity units electric power is also used as prime-mover.



POWERED FUNCTIONS → **Swing** – Hydraulic Motors **Propel** – Hydraulic Motors **Hoist** – Cylinders
Crowd - Cylinders **Bucket Wristing** - Cylinders **Bucket Dumping** - Cylinders

1 – Bucket	2 – Arm/Stick	3 – Boom	4 – Operator's Cab
5 – Upper Structure	6 – Swing Circle	7 – Undercarriage	8 – Crawler

General Features of Hydraulic Shovel

❖ Crawler mounted (two tracks)	❖ Fully revolving upper works
❖ All hydraulic drives	❖ Low profile machinery house

General Characteristics of Hydraulic Shovel

❖ Moderate ground pressure (0.9 to 1.97 kg/cm ²)	❖ High breakout forces through wristing of narrow bucket
❖ Medium overall mobility (about 1.5 mph)	❖ High bucket fill factors
❖ High gradeability	❖ High swing speeds (2.5 to 5 rpm)
❖ Excellent positioning capabilities (spin turns) with independent track drives	❖ Digging forces are not entirely limited by machine stability
❖ Good stability with good ground clearances	❖ Smooth, low shock dumping because dump action can be controlled
❖ Maximum versatility in bucket orientation for face penetration	❖ Loads directly into trucks with good placement of load
❖ Compact size and low weight	⇒ Operator position is safe; visibility is good for digging, but may be marginal for loading into larger trucks
❖ Low operator fatigue and moderate operator skill required	⇒ Medium service life (20,000 to 30,000 hours)
❖ No support equipment is required to maintain working area	⇒ Typical climatic temperature range (-39 to +49 degrees C)
❖ Can work in close quarters	➤ High initial cost
❖ Moderate dust generation	➤ Relatively high hydraulic maintenance

Applicability and Limitations of Electric Shovels and Hydraulic Excavators

Electric Shovel

These shovels are applied in benching operations in overburden/waste and coal/ore. The shovels excavate and load the bench materials onto off-highway trucks, or in some cases, mobile hoppers. These machines are simpler with high reliability and high sustained production capability. These are particularly suitable to more severe digging conditions.

However, these are not suited for the conditions where selective extraction of the material from the same bench is required.

Hydraulic Excavators

The hydraulic shovels are generally diesel powered with all hydraulic powered operating functions. These are primarily excavating and loading devices. While it can swing and/or propel to transport material short distances, it is used almost exclusively to excavate and load overburden/waste or coal/ore onto off-highway trucks, or in some cases, mobile hoppers. The diesel power makes these shovels more maneuverable that can be a huge advantage in a mine that wants to shift or balance out ore grade processed in a given day or week by moving the shovel around. The diesel power also allows these shovels to operate in remote mines far from an electric power plant.

These shovels are capable of much more precise digging and can be used efficiently where selective extraction of the material from the same bench is required, for example targeting a specific band of ore in a face. The smaller units may be used for utility work generally related to mine drainage systems.

Hydraulic back-hoes are more suitable for below grade digging, and this capability make them particularly suited to tasks as trenching or excavating underwater. These are utilized in mining when the conditions warrant keeping the equipment on the top of the bench surface.

4.1.1.2. Dumpers

Dumper/dump truck is a mobile piece of equipment for hauling/transporting material. It is often an integral part of material handling systems in surface mines either for transport of coal/ore from face to coal/ore stockpile or processing plant or for transport of overburden/waste to spoil dumps.

Development of Dumpers

The dumper/dump truck is thought to have been first conceived in the farms of late 19th century Western Europe. As early as 1905, the first motorized dumping vehicles were developed. The first motorized dump trucks in the United States were developed by small equipment companies such as Galion Buggy Co. and Lauth-Juergens among many others around 1910. Such companies flourished during World War-I due to massive wartime demand. Companies like Galion Buggy Co. continued to grow after the war by manufacturing a number of express bodies and some smaller dump bodies that could be easily installed on either stock or converted (heavy-duty suspension & drive train) Model T chassis prior to 1920. Galion and Wood Mfg., Co. built all of the dump bodies offered by Ford on their heavy-duty AA and BB chassis during the 1930s. Galion (now Galion Godwin Truck Body Co.) is the oldest known truck body manufacturer still in operation today (http://en.wikipedia.org/wiki/Dump_truck).

Prior to World War-II, the rail-road transport was very popular in surface mines as it was a bulk transport system that can handle any type of material over considerable distance economically. But the main drawbacks of this system are – requirement of very flat gradient (3% i.e. 1 in 33) and large turning radius, and the system is not flexible but rigidly engineered.

After the World War-II, truck transport system gradually became popular in surface mines due to the following reasons – it can also handle any type of material, its requirement of gradient (6% i.e. 1 in 16) is steeper than that of rail-road transport system, it can take sharp turns and the system is very much flexible. Because of the advantages offered by truck transport system over rail-road transport system and the increase in depth of operation, the rail-road transport system gradually became obsolete.

In the 1960s, with the continuous rise in diesel prices the truck transport system became more and more costly. In an effort to keep the unit transport cost within the economic range, higher and higher size of trucks were developed and used to take the advantage of scale of operation. Over the last few decades, the surface mining industry had witnessed the development of truck sizes from earlier 10 to 15 ton capacity (early 1950s) to 35 ton (1955-65) to 70 ton (late 1960s) to 150 ton (middle to late 1970s) to 240 ton (1985-95) to 320 ton (late 1990s) to 350 ton and then to 400 ton (Liebherr-T 282 C diesel electric) capacity (Walter G. Koellner et.al, 2004, “Recent Advances in Mining Haul Trucks”, IEEE Transactions on Industrial Electronics, Vol. 51, No. 2, April, pp. 321 – 329).

The power transmission and drive systems of trucks have also undergone considerable changes with the increase in capacity. Generally the trucks upto 85 ton capacity is having mechanical transmission system with torque converters and turbo chargers. Diesel–electric haul trucks with dc drives are built up to 240 ton capacity. For trucks having capacities more than 240 ton, high-power ac inverter-fed induction motor drives (similar to that are employed in locomotives) have been developed and are used.

Classification of Dumpers

Dumpers may be classified differently based on different criterion: –

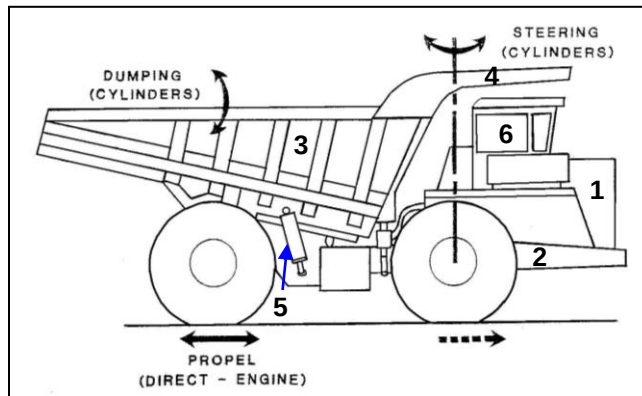
Dumping mode – Rear dump, bottom dump and side dump. In surface mining operation, rear dump trucks are most popular and side dump trucks are hardly used.

Frame type – Rigid frame / integrated chassis and articulated frame / tractor-trailer type. Dumpers of integrated chassis type are most common in surface mines.

Number of axles – Two axle type and three axle type. Dumpers with two axles are more popular in surface mines.

Apart from these, the dumpers may also be classified based on their carrying capacities (50 ton to 400 ton)) and drive (mechanical, electric) systems. Generally the higher capacity (≥ 120 ton) trucks have electric drive systems that are powered by a diesel generator installed on the truck.

General Features and Characteristics of Dumpers



Powered Functions → **Propel** – Direct - Engine **Steering** – Cylinders **Dumping** - Cylinders

1 – Engine	2 – Main Frame	3 – Body
4 – Canopy	5 – Body Lift Cylinder	6 – Cab

General Features

❖ Diesel Power	❖ Good traction with maximum weight applied to drive axle
❖ Good maneuverability	❖ Approximate service life – 20,000 hours for units upto 100 ton capacity and 30,000 hours for higher sizes
❖ Good ground clearance	

General Characteristics of Integrated Body Rear Dump Trucks

➤ Unit body construction	➤ Av. horsepower per ton of capacity – approx. 9.5
➤ Rugged body to withstand high loading impact	➤ Av. payload to net vehicle weight ratio – approx. 1.45
➤ High power and good gradeability	➤ Av. gross vehicle weight to horsepower ratio – approx. 350
➤ Rear axle-dual wheel drives	➤ Mechanical drive system upto 130 ton capacity
➤ Forward mounted engine	➤ Electrical wheel drive system from about 85 ton to 200 ton capacity
➤ Turning radius – about 1.1 to 1.2 times overall length	➤ Electrical axle drive system for capacities more than 200 ton

Applicability and Limitations of Dumpers

Dumpers are used for transporting coal/ore and/or overburden/waste from the face to the respective destinations. These can be used to haul coal/ore to stockpile or to a hopper, in practically any surface mine. Dumping to stockpile is generally done in shallow lifts. For feeding a hopper, a back-in hopper station is used for rear dump trucks and the bottom dump trucks move onto a drive over grizzly. The rear dump trucks, that carry the overburden/waste to spoil piles, dump the material following end-tipping method.

The truck transport system tends to be uneconomic when the haul distance from the working face to the dumping destination increases to more than 2 km. Also, with the increase in the depth of working, the haul distance between the pit floor and the surface increases that requires the placing and maintenance of longer haul roads thereby increasing the unit cost of transportation.

4.1.1.3. Cycle Time and Productivity Calculation for Shovel and dumper

Shovel Cycle Time

When a shovel loads a dumper, its working cycle consists of four components – (i) **digging** through the muck pile or the in-situ face (crowding, digging and hoisting) and loading the bucket, (ii) **forward swing** of the bucket over the dumper, (iii) **dumping** the bucket load onto the dumper and (iv) **return swing** of the bucket to the face at floor level to start the next cycle. The cycle time of a shovel is the total time required to complete a cycle consisting of the four above mentioned elements.

The standard cycle time is the time taken by the shovel to complete a cycle under ideal conditions for 90° swing and this is quoted by the manufacturers. However, the actual (average) cycle time for 90° swing is generally more than the standard one, and this is best estimated through time study in the field during actual working of the shovel. For this a time study over more than an hour is made and the total time is divided by the total number of cycles completed to get the actual cycle time.

Shovel Productivity

Shovel productivity is generally expressed in terms of m³ (bank)/hour for overburden and tonnes/hour for coal/ore or waste rock.

Theoretical productivity (Q_{th}) may be given as

$$Q_{th} = B_c \times B_f \times S_f \times \frac{3600}{t_c \times F_s} \text{ m}^3 \text{ (bank)/hour} = B_c \times B_f \times S_f \times H \times \frac{3600}{t_c \times F_s} \text{ m}^3 \text{ (bank)/year}$$

$$Q_{th} = B_c \times B_f \times S_f \times \rho \times \frac{3600}{t_c \times F_s} \text{ tonnes/hour} = B_c \times B_f \times S_f \times H \times \rho \times \frac{3600}{t_c \times F_s} \text{ tonnes/year}$$

However, the actual or effective productivity (Q_{eff}) is less than the theoretical productivity and is given by

$$Q_{eff} = B_c \times B_f \times S_f \times \frac{3600}{t_c \times F_s} \times E_s \text{ m}^3 \text{ (bank)/hour} = B_c \times B_f \times S_f \times H \times \frac{3600}{t_c \times F_s} \times E_s \text{ m}^3 \text{ (bank)/year}$$

$$Q_{eff} = B_c \times B_f \times S_f \times \rho \times \frac{3600}{t_c \times F_s} \times E_s \text{ tonnes/hour} = B_c \times B_f \times S_f \times H \times \rho \times \frac{3600}{t_c \times F_s} \times E_s \text{ tonnes/year}$$

Where,

B_c - bucket capacity, B_f - bucket fill factor, S_f - swell factor, H - working hours per year

P - in-situ specific gravity of coal/ore or waste rock, as the case may be

t_c - actual cycle time for 90° swing, F_s - factor of swing, E_s - overall utilization factor

Bucket fill factor (B_f) is a function of average material size, bucket size and the effective digging force. For the same average material size and the effective digging force, the bucket fill factor will increase with the increase of bucket size. Also, for the same bucket size and the effective digging force, the bucket fill factor will increase with the decrease of average material size.

Swell factor (S_f) is the ratio of in-situ volume to the fragmented volume for same mass of the material. Mathematically it may be expressed as 100 / (100 + % swell). Percentage swell is a function of degree of fragmentation and. As the degree of fragmentation increases, the percentage swell also increases limited to a maximum value defined by the material characteristics.

Factor of swing (F_s) is a cycle time correction factor to take into account the angle of swing that the shovel bucket has to make for loading the dumper. The swing angles and the corresponding F_s values are given in the following table.

Angle of swing (deg.) →	45	60	75	90	120	150	180
Swing Factor →	0.84	0.90	0.95	1.00	1.10	1.20	1.30

Overall utilization factor (E_s) is the ratio of actual working hours to the scheduled shift hours. Scheduled shift hours are the sum total of actual working hours, maintenance hours and the idle hours. In Indian surface mines the value of E_s is about 0.56.

Dumper Cycle Time

Dumper cycle time consists of six elements as shown below.

Dumper Cycle time = Spotting time + Loading time + Load travel time + Dumping time + Empty travel time + Waiting time at shovel end.

Spotting time is the time taken for a dumper to position itself beside the shovel for getting itself loaded. It generally varies between 0.5 to 1.0 min depending on the site conditions in case of single spotting condition and 0.0 min in case of double spotting condition. In single spotting condition the dumper can position only on one side of the shovel, whereas in double spotting condition the dumper can position itself on either side of the shovel.

Loading time is the actual time taken by the shovel to load a dumper

$$= (\text{No. of shovel bucket passes required to load the dumper}) \times (\text{shovel cycle time})$$

$$\text{Number of shovel bucket passes required} = \frac{\text{Effective dumper capacity}}{\text{Tonne per pass of shovel bucket}}$$

$$\begin{aligned}\text{Effective dumper capacity} &= \text{Rated dumper capacity} \times (\rho / 2.6), \rho < 2.6. \\ &= \text{Rated dumper capacity, otherwise}\end{aligned}$$

$$\text{Tonnes per pass of shovel bucket} = \text{Bucket capacity (m}^3\text{)} \times \text{bucket fill factor} \\ \times \text{swell factor} \times \text{in-situ specific gravity}$$

Number of shovel bucket passes required to load a dumper (NP1)

$$= \frac{\text{Effective dumper capacity}}{\text{Tonnes per pass of shovel bucket}}$$

Actual number of passes required (NP) = Integer part of NP1

$$\text{Loading time} = \text{Actual number of passes required} \times (\text{shovel cycle time (sec)} / 60) \text{ min}$$

$$\text{Load travel time} = \frac{\text{One way haul distance (km)}}{\text{Load travel speed (km / hour)}} \times 60 \text{ min}$$

Dumping time typically varies from 1.0 to 2.0 min depending on the dump site conditions.

$$\text{Empty travel time} = \frac{\text{One way haul distance (km)}}{\text{Empty travel speed (km / hour)}} \times 60 \text{ min}$$

Load travel speed and empty travel speed depend on conditions of the haul road and the gradients. Generally the load haul speed varies from 16 to 20 km / hour and empty haul speed varies from 24 to 30 km / hour.

To determine the dumper waiting time at the shovel end it is necessary to find out the actual number of dumpers required per shovel.

Number of dumpers required per shovel (ND1) is given by

$$\text{ND1} = \frac{\text{Spotting time} + \text{loading time} + \text{load travel time} + \text{dumping time} + \text{empty travel time}}{\text{Spotting time} + \text{loading time}}$$

Actual number of dumper required per shovel (ND) is given by

$$\begin{aligned}\text{ND} &= \text{ND1, if integer} \\ &= \text{Integer part of ND1} + 1, \text{ otherwise}\end{aligned}$$

$$\text{Waiting time at the shovel end} = (\text{ND} - \text{ND1}) \times (\text{Spotting time} + \text{Loading time})$$

Dumper Productivity

Dumper productivity is generally expressed in terms of tonne-km/hour both for overburden / waste rock and coal.

Theoretical productivity (DP_{th}) may be given as

$$DP_{th} = NP \times BL_s \times \frac{60}{CT_d} \times DIST \quad \text{tonne-km / hour}$$

Effective productivity (DP_{eff}) is given by

$$DP_{eff} = NP \times BL_s \times \frac{60}{CT_d} \times DIST \times E_d \quad \text{tonne-km / hour}$$

Where,

- NP - actual number of shovel bucket passes required to load a dumper
- BL_s - tonne per pass of shovel bucket
- CT_d - dumper cycle time (min)
- DIST - one way haul distance (km)
- E_d - overall utilization factor (value of E_d is about 0.50)

4.1.1.4. Estimation of Shovel, Dumper and other HEMM required in a Surface Coal Mine for a given Output

The steps to be followed for estimation of HEMM required in a surface coal mine using shovel-dumper combination for given production and other conditions are detailed below. For estimation of HEMM, the number of shovels and dumpers are calculated and the numbers of other equipment are estimated following certain norm.

Calculation for Number of Shovels and Dumpers

Let,

- P - required production of coal per year, tonne
- TH_c - average thickness of the coal seam
- SR - average stripping ratio, m^3/tonne
- H - total working hours per year (7200 for 3 shifts working)
- B_{co} - bucket capacity of shovel handling overburden, m^3
- B_{cc} - bucket capacity of coal shovel, m^3
- B_{fo} - bucket fill factor of overburden shovel (0.8 to 0.85)
- B_{fc} - bucket fill factor of coal shovel (0.8 to 0.85)
- S_o - % swell of over burden
- S_c - % swell of coal
- ρ_o - in-situ density of over burden (2.4 to 2.6), tonne/m^3
- ρ_c - in-situ density of coal (1.3 to 1.6) tonne/m^3
- E - overall utilization factor for shovels (0.56 for 3 shift working)
- F_s - swing factor for shovel (1 for 90 deg. swing)
- T_{co} - standard cycle time for overburden shovel, sec

Contd...

T_{cc}	- standard cycle time for coal shovel, sec
D_{co}	- capacity of overburden dumper, tonne
D_{cc}	- capacity of coal dumper, tonne
S_{to}	- spotting time for overburden dumper (0.2 to 1.0), min
S_{tc}	- spotting time for coal dumper (0.2 to 1.0), min
D_{to}	- dumping time for overburden dumper (1.0 to 2.0), min
D_{tc}	- dumping time for coal dumper (1.0 to 2.0), min
T_{lo}	- load travel speed for overburden dumper (20 to 30 kmph)
T_{lc}	- load travel speed for coal dumper (20 to 30 kmph).
T_{eo}	- empty travel speed for overburden dumper (1.5 times T_{lo})
T_{ec}	- empty travel speed for coal dumper (1.5 times T_{lc}).
$DIST_o$	- average distance of waste dump from overburden face, km
$DIST_c$	- average distance of coal stockyard from coal face, km

Calculation for Number of Coal Shovel

Total production of coal per year = P tonne.

Total workings hours per year = H

Effective production per hour per shovel,

$$Q_{ch} = B_{cc} \times B_{fc} \times \frac{100}{100 + S_c} \times \frac{3600}{t_{cc} \times F_s} \times \rho_c \times E \text{ tonne}$$

Effective production per year per shovel,

$$Q_{cy} = B_{cc} \times B_{fc} \times \frac{100}{100 + S_c} \times \frac{3600}{t_{cc} \times F_s} \times \rho_c \times H \times E \text{ tonne}$$

So, Number of working shovel required ($NS1_c$) = P / Q_{cy}

$$= \frac{P \times (100 + S_c) \times (t_{cc} \times F_s)}{B_{cc} \times B_{fc} \times 100 \times 3600 \times \rho_c \times H \times E}$$

Actual number of working shovel required for coal is given by -

$$\begin{aligned} NS_c &= NS1_c, \text{ if integer} \\ &= \text{Integer part of } NS1_c + 1, \text{ otherwise} \end{aligned}$$

Calculation for Number of Overburden Shovel

Total production of coal per year = P tonne.

Average stripping ration = SR m³/tonne.

So, Total excavation of overburden required per year = (P × SR) m³

Effective excavation capacity per hour per shovel,

$$Q_{oh} = B_{co} \times B_{fo} \times \frac{100}{100 + S_o} \times \frac{3600}{t_{co} \times F_s} \times E \text{ m}^3$$

Effective excavation capacity per year per shovel,

$$Q_{oy} = B_{co} \times B_{fo} \times \frac{100}{100 + S_o} \times \frac{3600}{t_{co} \times F_s} \times H \times E \text{ m}^3$$

So, Number of working shovel required (NS1_o) = (P × SR) / Q_{oy}

$$= \frac{P \times SR \times (100 + S_o) \times (t_{co} \times F_s)}{B_{co} \times B_{fo} \times 100 \times 3600 \times H \times E}$$

Actual number of working shovel required for overburden

$$\begin{aligned} NS_o &= NS1_o, \text{ if integer} \\ &= \text{Integer part of } NS1_o + 1, \text{ otherwise} \end{aligned}$$

Calculation for Number of Dumpers for Coal

Dumper Cycle time = Spotting time + Loading time + Load travel time + Dumping time + Empty travel time + Waiting time at shovel end.

Loading time is the actual time taken by the shovel to load a dumper
= (No of shovel bucket passes required to load the dumper) × (shovel cycle time)

$$\text{Number of shovel bucket passes required} = \frac{\text{Effective dumper capacity}}{\text{Tonne per pass of shovel bucket}}$$

$$\begin{aligned} \text{Effective dumper capacity} &= \text{Rated dumper capacity} \times (\rho/2.6), \quad \rho < 2.6. \\ &= \text{Rated dumper capacity, otherwise} \end{aligned}$$

$$\begin{aligned} \text{Tonnes per pass of shovel bucket} &= \text{Bucket capacity (m}^3\text{)} \times (100 / (100 + \% \text{ Swell})) \\ &\quad \times \text{fill factor} \times \text{in-situ density (tonne/m}^3\text{)} \end{aligned}$$

In this case,

$$\text{Effective dumper capacity} = (D_{cc} \times \rho_c) / 2.6$$

$$\text{Tonnes per pass of shovel bucket} = B_{cc} \times B_{fc} \times \rho_c \times (100 / (100 + S_c))$$

$$\begin{aligned} \text{Number of shovel bucket passes required to load a dumper (NP1)} \\ &= (D_{cc} \times \rho_c \times (100 + S_c)) / (2.6 \times B_{cc} \times B_{fc} \times \rho_c \times 100) \end{aligned}$$

Actual number of passes required, NP_c = Integer part of NP1

$$\text{So, the dumper loading time} = NP_c \times t_{cc} / 60 \text{ (min)}$$

$$\text{Load travel time} = (DIST_c / T_{lc}) \times 60 \text{ (min)}$$

$$\text{Empty travel time} = (DIST_c / T_{ec}) \times 60 \text{ (min)}$$

The number of dumpers required per shovel (ND1_c)

$$\begin{aligned} &= \frac{\text{Spotting time} + \text{Loading time} + \text{Load travel time} + \text{Dumping time} + \text{Empty travel time}}{\text{Spotting time} + \text{Loading time}} \\ &= \frac{[S_{tc} + ((NP_c \times t_{cc})/60) + ((DIST_c \times 60)/T_{lc}) + D_{tc} + ((DIST_c \times 60)/T_{ec})]}{[S_{tc} + ((NP_c \times t_{cc})/60)]} \end{aligned}$$

Actual number of working dumper required per working coal shovel,

$$\begin{aligned} ND_c &= ND1_c, \text{ if integer} \\ &= \text{Integer part of } ND1_c + 1, \text{ otherwise} \end{aligned}$$

So, total number of working dumpers for coal = NS_c × ND_c

Calculation for Number of Dumpers for Overburden

Effective dumper capacity = D_{co}

Tonnes per pass of shovel bucket = $B_{co} \times B_{fo} \times \rho_o \times 100 / (100 + S_o)$

Number of shovel bucket passes required to load a dumper (NP2)
 $= (D_{co} \times (100 + S_o)) / (B_{co} \times B_{fo} \times \rho_o \times 100)$

Actual number of passes required, NP_o = Integer part of NP2.

So, the dumper loading time = $(NP_o \times t_{co}) / 60$ (min)

Load travel time = $(DIST_o / T_{lo}) \times 60$ (min)

Empty travel time = $(DIST_o / T_{eo}) \times 60$ (min)

So, the number of dumpers required per shovel ($ND1_o$)

$$= \frac{[S_{to} + ((NP_o \times t_{co})/60) + ((DIST_o \times 60)/T_{lo}) + D_{to} + ((DIST_o \times 60)/T_{eo})]}{[S_{to} + ((NP_o \times t_{co})/60)]}$$

Actual number of working dumpers required per working overburden shovel,

$$ND_o = ND1_o \text{ (if integer)} \\ = \text{Integer part of } ND1_o + 1 \text{ (otherwise)}$$

So, the number of working dumpers for overburden = $NS_o \times ND_o$

Estimation of Other HEMM

Number of working drills required = Number of working shovels

Number of working dozers required = 0.5 per working shovel + 1 to 1.5 for sundry jobs
+ 1 or 2 at coal stockyard + 1 or 2 at waste dump

Number of working front-end loaders required = 1 or 2 for supporting jobs at the pit + 1 to 3
(based on production level) at the coal stockyard.

Number of working motor graders required = 1 or 2 for road maintenance.

Number of cranes required = 1 or more depending on production level

Number of water sprinklers required = 1 or more based on production level

Numerical Example

Estimate the HEMM required in a surface mine producing 4.5 million tonne of coal per year and being worked by shovel dumper combination. The average seam thickness is 10m and the average stripping ratio is 2.5 m³/tonne.

Given Conditions:

Production of coal per year = 4.5×10^6 tonne

Average stripping ratio = 2.5 m³/tonne

Coal seam thickness = 10 m.

Assumptions:

Total available hours/year = $300 \times 3 \times 8 = 7200$

Overall utilization of available hours for shovels = 56%

Bank density of coal = 1.4 te/m³

Coal shovel bucket capacity = 5m^3

Overburden shovel bucket capacity = 5m^3

Coal shovel fill factor = 0.85

Overburden shovel fill factor = 0.8

% Swell of coal = % Swell of overburden = 20%

Bank density of overburden = 2.6 te/m^3

Angle of swing for coal shovel and overburden shovel = 90 deg.

Standard cycle time of shovel (both for coal and overburden) = 30 sec.

Rated Capacity of dumpers (both for coal and overburden) = 50 te.

Spotting time for dumpers (both for coal and overburden) = 0.3 min.

Dumping time for dumpers (both for coal overburden) = 1.5 min.

Load travel speed for dumpers (both for coal overburden) = 20 kmph.

Empty travel speed for dumpers (both for coal and overburden) = 30 kmph.

Haul distance (one way) both for coal and overburden = 1 km.

Calculation

Stripping ratio =
$$\frac{\text{Average thickness of overburden}}{\text{Average thickness of coal seam} \times \text{in-situ density of coal}}$$

So, Average Thickness of overburden = $2.5 \times 10 \times 1.4\text{ m.} = 35\text{ m.}$

This may be handled by 3 benches of height 12 m, 12m and 11 m.

Number of coal shovel

Required production per year = $(4.5 \times 10^6)\text{ tonne}$

Effective production per year per shovel

$$= (5 \times 0.85 \times 100 \times 3600 \times 1.4 \times 7200 \times 0.56) / ((100 + 20) \times 30)\text{ te} = 2,399,040\text{ te}$$

Number of working shovels required = $(4.5 \times 10^6) / 2399040 = 1.87\ldots$

Actual number of working shovels required = 2

Number of overburden shovels

Required excavation capacity per year = $4.5 \times 10^6 \times 2.5\text{ m}^3$

Effective excavation capacity per year per shovel

$$= (5 \times 0.8 \times 100 \times 3600 \times 7200 \times 0.56) / ((100 + 20) \times 30) = 1,612,800\text{ m}^3$$

Number of working shovels required = $(4.5 \times 10^6 \times 2.5) / 1,612,800 = 6.97\ldots$

Actual number of working shovels required = 7

Number of Dumpers for coal

Effective dumper capacity = $(50 \times 1.4) / 2.6\text{ te} = 27\text{ te}$

Tonnes per pass of shovel bucket = $(5 \times 0.85 \times 1.4 \times 100) / (100 + 20) = 4.958\ldots$

Number of passes required to load the dumper = $(27 / 4.96) = 5.44\ldots$

Actual number of pass = 5 $\Rightarrow$ Dumper Loading time = $5 \times 30\text{ sec} = 2.5\text{ min}$

Spotting time = 0.3 min

Load travel time = $(1 \times 60) / 20 \text{ min} = 3 \text{ min}$

Dumping time = 1.5 min

Empty travel time = $(1 \times 60) / 30 \text{ min} = 2 \text{ min}$

Total number of dumpers required per shovel = $(0.3 + 2.5 + 3 + 1.5 + 2) / (0.3 + 2.5)$
 $= 9.3 / 2.8 = 3.32...$

Actual number of dumpers required per shovel = 4

Waiting time for each dumper = $(4 \times 2.8) - 9.3 \text{ min} = 11.2 - 9.3 \text{ min} = 1.9 \text{ min}$

Total number of dumpers required for coal = $2 \times 4 = 8$

Number of Dumpers for overburden

Effective dumper capacity = 50 te.

Tonnes per pass of shovel bucket = $(5 \times 0.8 \times 2.6 \times 100) / (100 + 20) = 8.666...$

Number of passes required to load the dumper = $50 / 8.67 = 5.76...$

Actual number of pass = 5 => Dumper loading time = $5 \times 30 \text{ sec} = 2.5 \text{ min}$.

Spotting time = 0.3 min.

Load travel time = 3 min.

Dumping time = 1.5 min.

Empty travel time = 2 min.

The number of dumpers required per shovel = $(0.3 + 2.5 + 3 + 1.5 + 2) / (0.3 + 2.5)$
 $= 9.3 / 2.8 = 3.32...$

Actual number of dumpers required per shovel = 4

Waiting time for each dumper = $(4 \times 2.8) - 9.3 \text{ min} = 11.2 - 9.3 \text{ min} = 1.9 \text{ min}$

Total number of dumpers required for overburden = $7 \times 4 = 28$.

Statement of HEMM Required in the Mine

Sl. No.	Item	Capacity	Qty.		Remarks
			Working	Total	
1.	Coal shovel	5.0 m ³	2	3	Spare-1
2.	Dumper for coal shovels	50 te.	8	12	Spare-4
3.	Drill for coal faces	150 mm.	2	3	Spare-1
4.	Overburden shovel	5.0 m ³	7	9	Spare-2
5.	Dumper for overburden shovels	50 te.	28	42	Spare-14
6.	Drill for overburden faces	250 mm.	7	9	Spare-2
7.	Dozer	250/350 hp	9	11	Spare-2
8.	Front end loader	6.0 m ³	3	4	Spare-1
9.	Motor grader	-	2	3	Spare-1
10.	Crane		2	3	Spare-1
11.	Water sprinkler		2	3	Spare-1

4.1.1.5. Estimation of Shovel, Dumper and other HEMM required in a Surface Iron Ore Mine for a given Output

The steps to be followed for estimation of HEMM required in a surface iron ore mine, or any other mine where both ore and waste are mined in same bench, using shovel-dumper combination for given production and other conditions are detailed below. For estimation of HEMM, the number of shovels and dumpers are calculated and the numbers of other equipment are estimated following certain norm.

Calculation for Number of Shovels and Dumpers

Let,

- P - required production of ore per year, tonne
- SR - average stripping ratio, tonne/tonne
- H - total working hours per year (7200 for 3 shifts working)
- B_c - bucket capacity of shovel, m³
- B_f - bucket fill factor of shovel (0.8 to 0.85)
- S - % swell of ore and waste
- ρ_o - in-situ density of ore, tonne/m³
- ρ_w - in-situ density of waste, tonne/m³
- E - overall utilization factor for shovels (0.56 for 3 shift working)
- F_s - swing factor for shovel (1 for 90 deg. swing)
- T_c - standard cycle time for shovel, sec
- D_c - capacity of dumper, tonne
- S_t - spotting time for dumper (0.2 to 1.0), min
- D_t - dumping time for dumper (1.0 to 2.0), min
- T_l - load travel speed for dumper (20 to 30 kmph)
- T_e - empty travel speed for dumper (1.5 times T_l)
- DIST - average haul distance for ore and waste, km

Calculation for Number of Shovel

Total production of ore per year = P tonne

Total production of waste per year = P × SR tonne

Total production of ore and waste per year = P × (1 + SR) tonne

Total volume of ore per year = P / ρ_o m³

Total volume of waste per year = (P × SR) / ρ_w m³

Total volume of ore and waste per year = P [(1/ρ_o) + (SR / ρ_w)] m³

Average density of composite ore and waste, ρ_{av} = [P × (1 + SR)] / P [(1/ρ_o) + (SR / ρ_w)] te/ m³
= (1 + SR) / [(1/ρ_o) + (SR / ρ_w)] te/ m³ = [(1 + SR) × ρ_o × ρ_w] / (ρ_w + SR × ρ_o) te/ m³

Total workings hours per year = H

Effective production per hour per shovel,

$$Q_{ch} = B_c \times B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \times \rho_{av} \times E \text{ tonne}$$

Effective production per year per shovel,

$$Q_{cy} = B_c \times B_f \times \frac{100}{100 + S} \times \frac{3600}{t_c \times F_s} \times \rho_{av} \times H \times E \text{ tonne}$$

So, Number of working shovel required (NS1) = $P \times (1 + SR) / Q_{cy}$

$$= \frac{P \times (1 + SR) \times (100 + S) \times (t_c \times F_s)}{B_c \times B_f \times 100 \times 3600 \times \rho_{av} \times H \times E}$$

Actual number of working shovel required is given by -

$$\begin{aligned} NS &= NS1, \text{ if integer} \\ &= \text{Integer part of } NS1 + 1, \text{ otherwise} \end{aligned}$$

Calculation for Number of Dumpers

Dumper Cycle time = Spotting time + Loading time + Load travel time + Dumping time + Empty travel time + Waiting time at shovel end.

Loading time is the actual time taken by the shovel to load a dumper

$$= (\text{No of shovel bucket passes required to load the dumper}) \times (\text{shovel cycle time})$$

$$\text{Number of shovel bucket passes required} = \frac{\text{Effective dumper capacity}}{\text{Tonne per pass of shovel bucket}}$$

$$\begin{aligned} \text{Effective dumper capacity} &= \text{Rated dumper capacity} \times (\rho/2.6), \quad \rho < 2.6. \\ &= \text{Rated dumper capacity}, \text{ otherwise} \end{aligned}$$

$$\begin{aligned} \text{Tonnes per pass of shovel bucket} &= \text{Bucket capacity (m}^3\text{)} \times (100 / (100 + \% \text{ Swell})) \\ &\quad \times \text{fill factor} \times \text{in-situ density (tonne/m}^3\text{)} \end{aligned}$$

In this case,

$$\text{Effective dumper capacity} = D_c$$

$$\text{Tonnes per pass of shovel bucket} = B_c \times B_f \times \rho_{av} \times (100/(100 + S))$$

Number of shovel bucket passes required to load a dumper (NP1)

$$= (D_c \times (100 + S)) / (B_c \times B_f \times \rho_{av} \times 100)$$

Actual number of passes required, NP = Integer part of NP1

So, the dumper loading time = $NP \times t_c / 60$ (min)

Load travel time = $(DIST / T_l) \times 60$ (min)

Empty travel time = $(DIST / T_e) \times 60$ (min)

The number of dumpers required per shovel (ND1)

$$\begin{aligned} &= \frac{\text{Spotting time} + \text{Loading time} + \text{Load travel time} + \text{Dumping time} + \text{Empty travel time}}{\text{Spotting time} + \text{Loading time}} \\ &= \frac{[S_t + ((NP \times t_c)/60) + ((DIST \times 60)/T_l) + D_t + ((DIST \times 60)/T_e)]}{[S_t + ((NP \times t_c)/60)]} \end{aligned}$$

Actual number of working dumper required per working coal shovel,

$$\begin{aligned} ND &= ND1, \text{ if integer} \\ &= \text{Integer part of } ND1 + 1, \text{ otherwise} \end{aligned}$$

So, total number of working dumpers = $NS \times ND$

Estimation of Other HEMM

Number of working drills required = Number of working shovels

Number of working dozers required = 0.5 per working shovel + 1 to 1.5 for sundry jobs + 1 or 2 at crusher + 1 or 2 at ore stockpile area + 1 or 2 at waste dump

Number of working front-end loaders required = 1 or 2 for supporting jobs at the pit + 1 to 3 (based on production level) at the ore stockpile area

Number of working motor graders required = 1 or 2 for road maintenance.

Number of cranes required = 1 or more depending on production level

Number of water sprinklers required = 1 or more based on production level

Numerical Example

Estimate the HEMM required in a surface Iron Ore mine for producing 6.0 million tonne of ore per year. The average stripping ratio is 0.3 tonne/tonne. The mine will be worked by shovel dumper combination with a bench height of 12m.

Given Conditions:

Production of ore per year = 6.0×10^6 tonne

Average stripping ratio = 0.3 tonne/tonne

Bench height = 12 m.

Assumptions:

Total available hours/year = $300 \times 3 \times 8 = 7200$

Overall utilization of available hours for shovels = 56%

Bank density of ore = 3.6 te/m^3

Bank density of waste = 2.8 te/m^3

Shovel bucket capacity = 5 m^3 .

Shovel fill factor = 0.8.

% Swell for ore and waste = 25%.

Angle of swing for shovel = 90 deg.

Standard cycle time of shovel = 30 sec.

Rated Capacity of dumpers = 50 te.

Spotting time for dumpers = 0.3 min.

Dumping time for dumpers = 1.5 min.

Load travel speed for dumpers = 20 kmph.

Empty travel speed for dumpers = 30 kmph.

Haul distance (one way) both for ore and waste = 1 km.

Calculation

Production of ore per year = 6.0×10^6 tonne

Handling of waste per year = $0.3 \times 6.0 \times 10^6$ tonne = 1.8×10^6 tonne

Total material to be handled per year = $(6.0 + 1.8) \times 10^6$ tonne = 7.8×10^6 tonne

Bank density of ore = 3.6 te/m^3 Bank density of waste = 2.8 te/m^3

Average density of composite ore and waste, $\rho_{av} = [(1 + SR) \times \rho_o \times \rho_w] / (\rho_w + SR \times \rho_o) \text{ te/ m}^3$

Average density of composite ore and waste,

$$\begin{aligned}\rho_{av} &= [(1 + 0.3) \times 3.6 \times 2.8] / (2.8 + 0.3 \times 3.6) \text{ te/ m}^3 = (1.3 \times 3.6 \times 2.8) / (2.8 + 1.08) \text{ te/ m}^3 \\ &= (13.104 / 3.88) \text{ te/ m}^3 = 3.377... \text{ te/ m}^3\end{aligned}$$

Number of shovel

Required handling of total material (ore and waste) per year = (7.8×10^6) tonne

Average density of composite ore and waste = $(13.104 / 3.88) \text{ te/ m}^3$

Effective production per year per shovel

$$\begin{aligned}&= (5 \times 0.8 \times 100 \times 3600 \times 13.104 \times 7200 \times 0.56) / ((100 + 25) \times 3.88 \times 30) \text{ te} \\ &= 76,082,872,320 / 14,550 \text{ te} = 5,229,063.39 \text{ te}\end{aligned}$$

Number of working shovels required = $(7.8 \times 10^6) / 5,229,063.39 = 1.49...$

Actual number of working shovels required = 2

Number of Dumpers

Effective dumper capacity = 50 te, as $\rho_{av} > 2.6 \text{ te/ m}^3$

Tonnes per pass of shovel bucket = $(5 \times 0.8 \times 13.104 \times 100) / ((100 + 25) \times 3.88) = 10.807...$

Number of passes required to load the dumper = $(50 / 10.807) = 4.62...$

Actual number of pass = 4 => Dumper Loading time = $4 \times 30 \text{ sec} = 2.0 \text{ min}$.

Spotting time = 0.3 min.

Load travel time = $(1 \times 60) / 20 \text{ min} = 3 \text{ min}$

Dumping time = 1.5 min

Empty travel time = $(1 \times 60) / 30 \text{ min} = 2 \text{ min}$

Total number of dumpers required per shovel = $(0.3 + 2 + 3 + 1.5 + 2) / (0.3 + 2)$
 $= 8.8 / 2.3 = 3.82...$

Actual number of dumpers required per shovel = 4

Waiting time for each dumper = $(4 \times 2.3) - 8.8 \text{ min} = 9.2 - 8.8 \text{ min} = 0.4 \text{ min}$

Total number of dumpers required = $2 \times 4 = 8$

Statement of HEMM Required in the Mine

Sl. No.	Item	Capacity	Qty.		Remarks
			Working	Total	
1.	Shovel	5.0 m ³	2	2	Spare included
2.	Dumper	50 te.	8	12	Spare-4
3.	Drill	225 mm.	2	2	Spare included
4.	Dozer	250/350 hp	5	6	Spare-1
5.	Front end loader	3.0 m ³	3	4	Spare-1
6.	Motor grader	-	2	3	Spare-1
7.	Crane	-	2	3	Spare-1
8.	Water sprinkler	-	2	3	Spare-1

4.1.1.6. Dependence of Shovel Productivity on Dumper Capacity

Let a shovel of bucket capacity B_c is loading two dumpers of significantly different capacity.

The shovel cycle time = t_c min

Effective bucket load per pass of shovel = $B_l = B_c \times B_f \times (100/(100+S)) \times \rho$ te.

No. of passes required to fill the smaller capacity dumper = n

No. of passes required to fill the larger capacity dumper = $n + m$

For **single spotting condition**, spotting time for both the dumpers = t_s min

For the smaller dumper,

No. of pass to fill the dumper = $n \Rightarrow$ Total time to fill the dumper = $n \times t_c$ min

Effective total time to fill the dumper = $(n \times t_c) + t_s$ min

No. of dumpers filled per hour = $60 / ((n \times t_c) + t_s)$

Effective hourly shovel productivity, $P_{sd} = (60 \times n \times B_l) / ((n \times t_c) + t_s) = (60 \times B_l) / (t_c + (t_s / n))$

Similarly, for the larger dumper,

Effective hourly shovel productivity, $P_{ld} = (60 \times (n + m) \times B_l) / (((n + m) \times t_c) + t_s)$
 $= (60 \times B_l) / [t_c + (t_s / (n + m))]$

As $(n + m) > n$, $P_{ld} > P_{sd}$

For **double spotting condition**, spotting time for both the dumpers = 0 min

Under this condition,

The effective hourly shovel productivity corresponding to both the dumpers = $(60 \times B_l) / t_c$

So for a given shovel, its effective productivity will increase with significant increase in dumper capacity in single spotting condition. However, in double spotting condition the effective productivity of a shovel becomes independent of the dumper capacity.

Dependence of Dumper Productivity on Shovel Capacity

Let a dumper is being loaded by two shovels of significantly different capacity.

The cycle time for smaller shovel = t_c min

The cycle time for larger shovel = $(t_c + \Delta t_c)$ min

Effective bucket load per pass of smaller capacity shovel = $B_{l1} = B_{c1} \times B_{f1} \times (100/(100+S)) \times \rho \text{ te}$.

Effective bucket load per pass of larger capacity shovel = $B_{l2} = B_{c2} \times B_{f2} \times (100/(100+S)) \times \rho \text{ te}$.

No. of passes required to fill the dumper by smaller capacity shovel = n

No. of passes required to fill the dumper by larger capacity shovel = $(n - m)$

So, $n \times B_{l1} \approx (n - m) \times B_{l2}$

The loading time required for the dumper by smaller capacity shovel, $t_{l1} = (n \times t_c)$ min

The loading time required for the dumper by larger capacity shovel, $t_{l2} = (n - m) \times (t_c + \Delta t_c)$ min
 $= ((n - m) \times t_c) \text{ min} + ((n - m) \times \Delta t_c) \text{ min}$

As $\Delta t_c \ll t_c$, $((n - m) \times t_c) < (n \times t_c) \Rightarrow t_{l2} < t_{l1}$

Total dumper cycle time when loaded by smaller shovel, $t_{dc1} = (t_s + t_{l1} + t_{lt} + t_d + t_{et})$ min

Total dumper cycle time when loaded by larger shovel, $t_{dc2} = (t_s + t_{l2} + t_{lt} + t_d + t_{et})$ min

As $t_{l2} < t_{l1}$, $t_{dc2} < t_{dc1}$

Effective hourly dumper productivity when loaded by smaller shovel, $P_{s1} = (60 \times n \times B_{l1}) / t_{dc1}$

Effective hourly dumper productivity when loaded by larger shovel, $P_{s2} = (60 \times (n - m) \times B_{l1}) / t_{dc2}$
 $= (60 \times n \times B_{l1}) / t_{dc2}$

As $t_{dc2} < t_{dc1}$, $P_{s1} < P_{s2}$

This is true for both the single and double spotting condition. Hence, the effective dumper productivity increases with the significant increase in shovel bucket capacity irrespective of spotting condition.

4.1.1.7. Dynamic Allocation of Trucks / Truck Dispatching System

4.1.1.8. Method of Work using Shovel–Dumper Combination

The shovel-dumper combination can be applied, from the technology point of view, for extraction of any type of subsurface deposits (bedded, vein, pipe, massive) and hilly deposits by surface mining.

4.1.1.8.1. Sub-Surface Bedded Deposits

The methods of work for bedded deposits have been discussed below taking coal seam/seams (bedded deposit) as example.

When a surface coal mine is worked by shovel–dumper combination, the system of the bench formation and the characteristic features of the working method vary considerably depending on the gradient of the coal seam/seams.

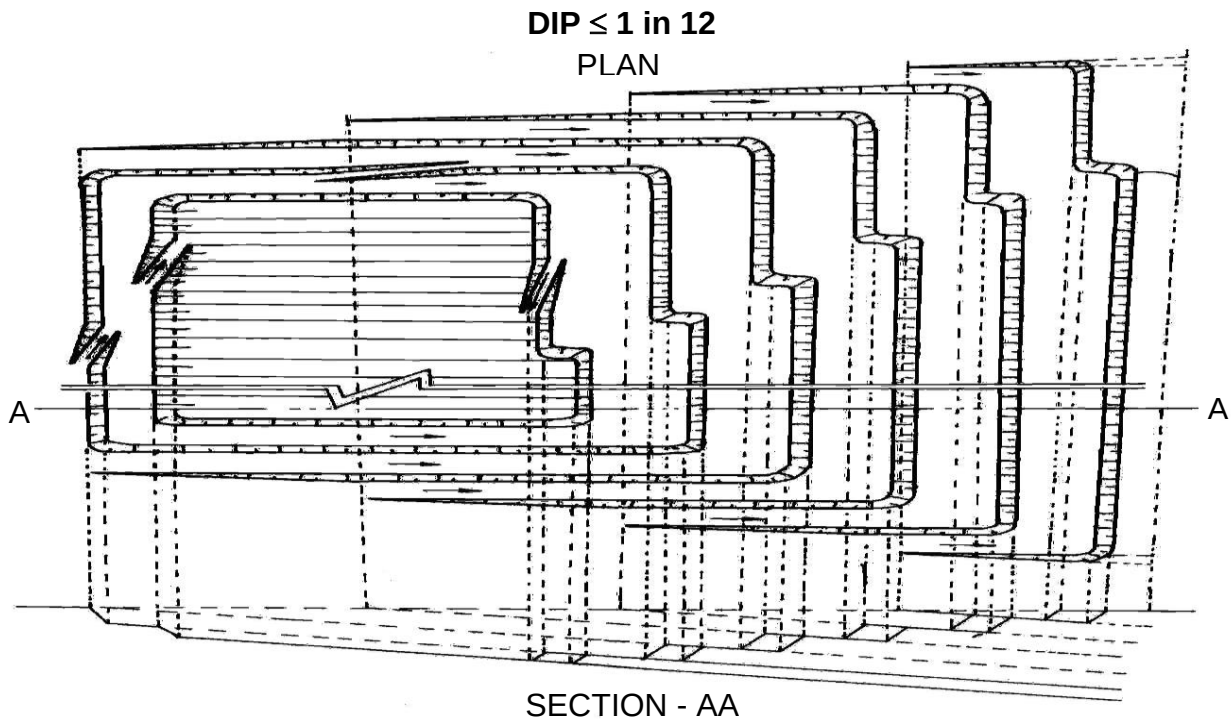
From the point of view of variation in working method, the coal seams may be classified into four groups – horizontal and flat seams ($\text{dip} \leq 1$ in 12), gently dipping seams ($1 \text{ in } 12 < \text{dip} \leq 1 \text{ in } 8$), moderately dipping seams ($1 \text{ in } 8 < \text{dip} \leq 1 \text{ in } 3$) and steeply dipping seams ($\text{dip} > 1 \text{ in } 3$).

Horizontal and Flat Seams

In case of seam/seams upto a gradient of not more than 1 in 12, the box-cut is established on the rise most portion of the property and from there the working benches are formed/developed. As the pit advances towards the dip side more and more benches in overburden are added on the top of the earlier bench for opening up the coal seam/seams that is extracted in single or multiple benches (as the case may be) subsequently.

All the overburden benches are made parallel to the coal bench/benches. The overburden benches and the coal bench/benches are distinctly separate, and all the benches move in the same direction towards the dip. Finally, when the overburden ratio reaches a limiting value (no profit no loss) the workings are stopped. Transportation of coal to coal depot and the overburden excavated in the initial period to outside spoil dump is done through the established haul road ramp system.

As the floor of the seam is flat, the overburden extracted may be backfilled in the de-coaled zone simultaneously once sufficient de-coaled area is available. While backfilling a clear distance of 50 to 80 meter should be maintained between the toe of the coal bench and that of the backfilled overburden bench at the de-coaled floor level. The sequence of rock strata in the highwall should also be maintained in the backfilled dump.

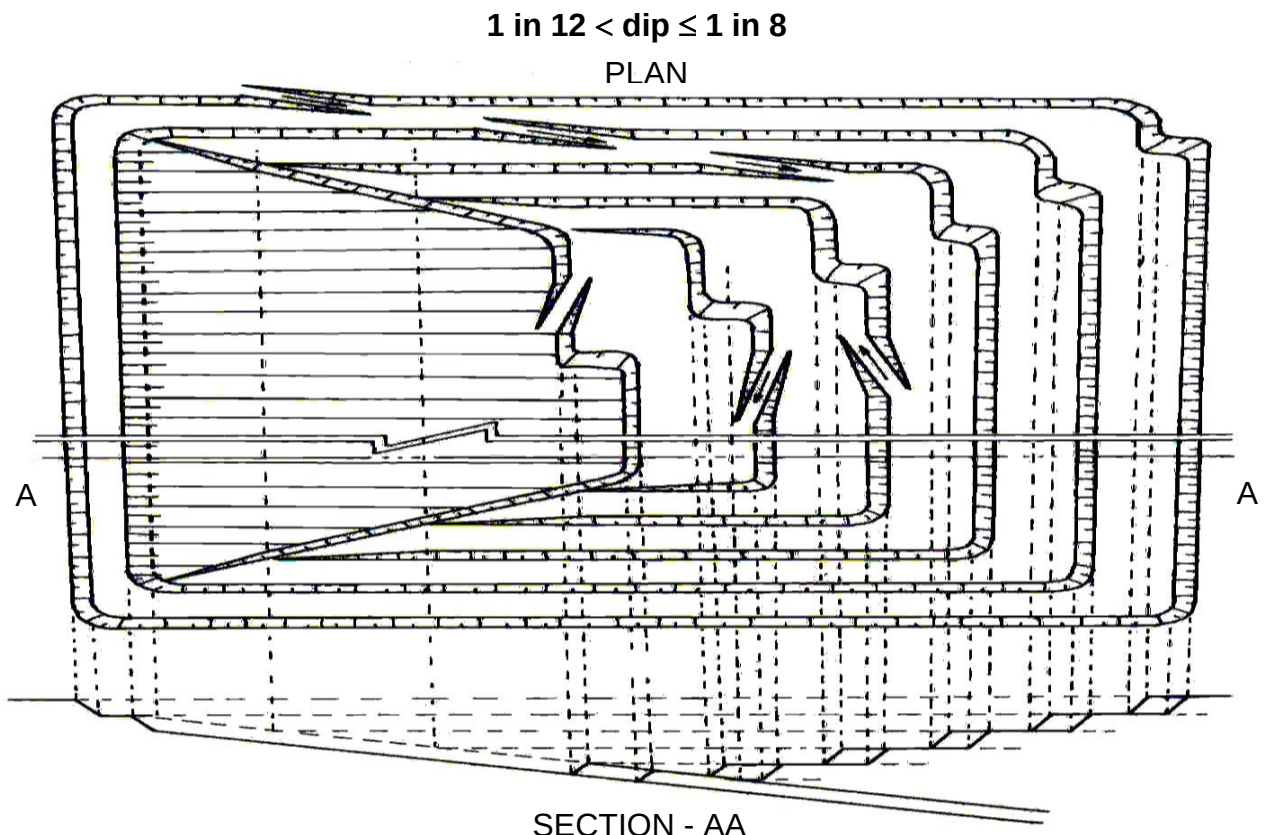


Gently Dipping Seams

In case of mining gently inclined ($1 \text{ in } 12 < \text{dip} \leq 1 \text{ in } 8$) coal seams, the method of work and the characteristic features are more or less same with the method of working for flat seams. The overburden benches and the coal bench/benches are distinctly separate, and all the benches move in the same direction towards the dip. Finally, when the overburden ratio reaches a limiting value (no profit no loss) the workings are stopped. Transportation of coal to coal depot and the portion of the overburden excavated to be emplaced in outside spoil dump is done through the established haul road ramp system.

However, in this case the overburden benches are made horizontal and as the pit advances towards dip side, more and more overburden benches are added at the bottom of the earlier bench for opening up the coal seam for its subsequent extraction.

Also a considerable portion of the overburden extracted may be backfilled in the de-coaled zone simultaneously once sufficient de-coaled area is available. While backfilling a clear distance of 50 to 80 meter should be maintained between the toe of the coal bench and that of the backfilled overburden bench at the de-coaled floor level. The sequence of rock strata in the highwall should also be maintained in the backfilled dump.



Moderately Dipping Seams

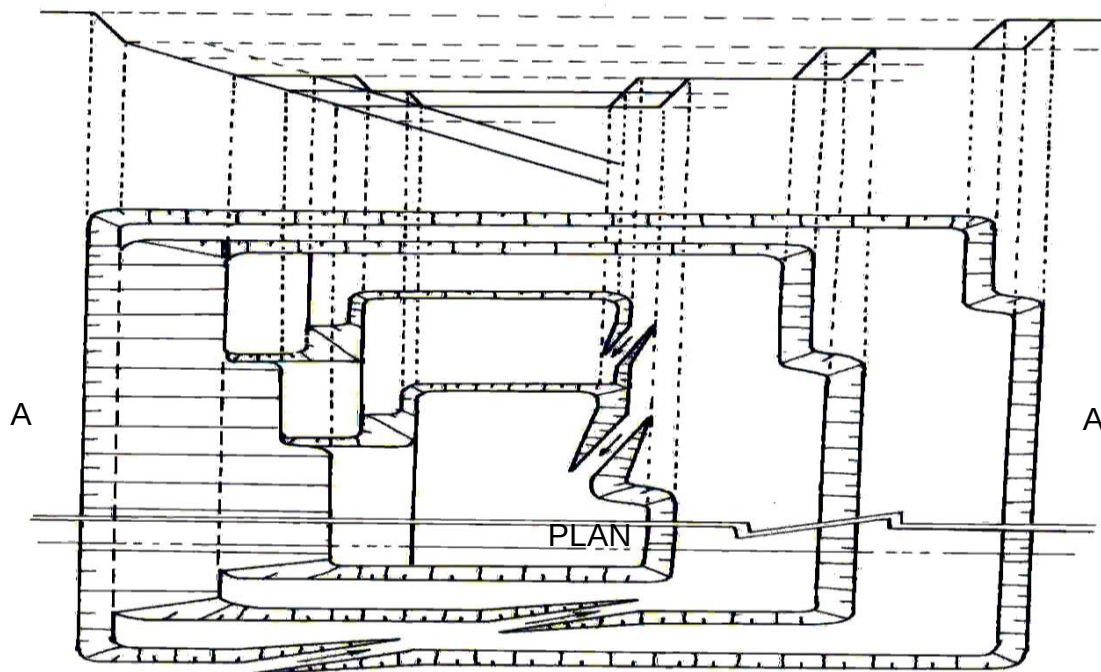
The method of mining of moderately inclined ($1 \text{ in } 8 < \text{dip} \leq 1 \text{ in } 3$) seams is totally different from that of flat or gently inclined seams. In this case, a box-cut is first made on the hang wall side of the seam. The cut is extended in both the directions, towards the coal seam and the hang wall,

and also in the strike direction. When the cut reaches the coal seam, the seam is released by breaking and dozing the triangular portion of overburden attached to it followed by extraction of the portion of the coal seam released upto the footwall contact. The process is repeated till the final pit depth is reached.

In this method the pit primarily advances towards the depth (vertical advance), and the advance in the sidewalls and the hang wall side in form of benches is made for stability of pit walls and for safety. The coal face and the overburden bench faces move in opposite direction. Both the coal and the overburden in the lowermost overburden bench are excavated standing on the same pit floor, i.e. the benches are not distinctly separate. All the benches are made horizontal and the footwall contact is generally not kept benched unless it is incompetent. The coal is transported to coal depot and the overburden is transported to outside spoil dump through the established haul road ramp system.

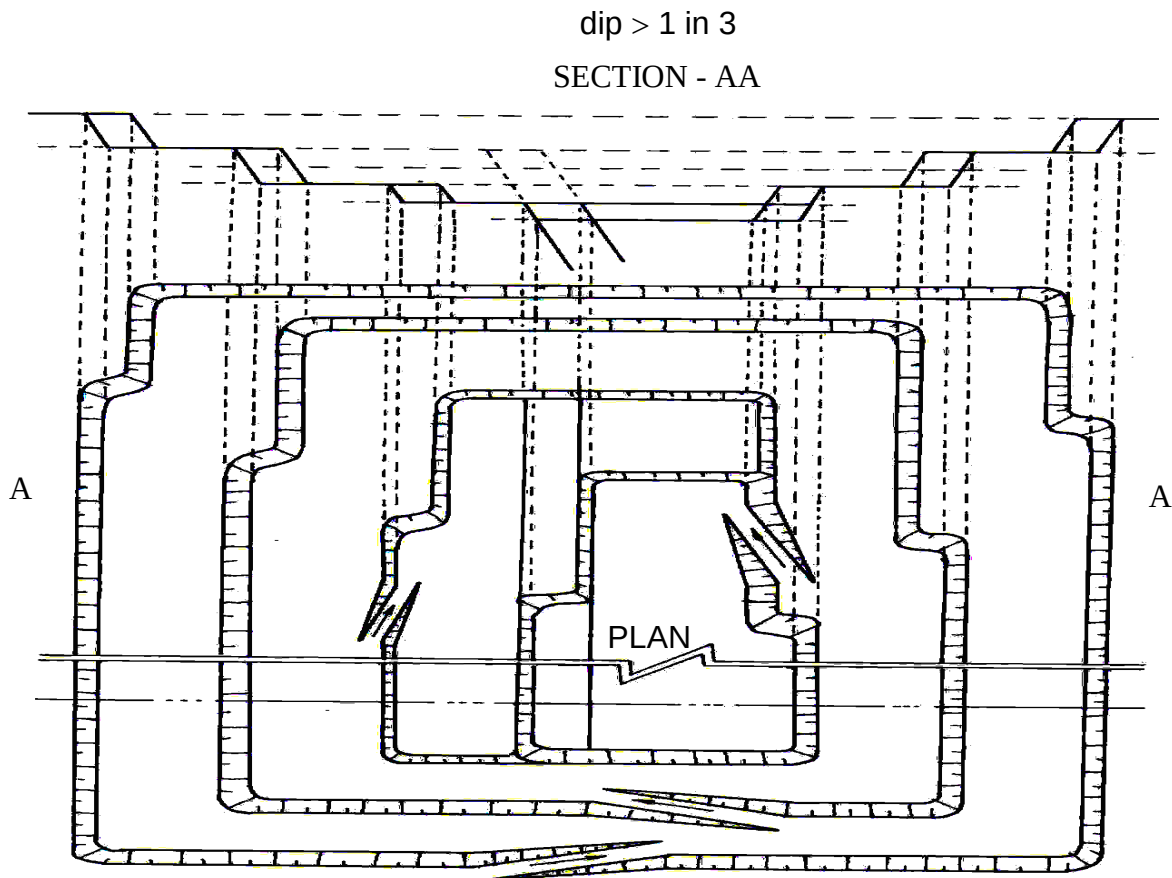
$1 \text{ in } 8 < \text{dip} \leq 1 \text{ in } 3$

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Steeply Dipping Seams

The method of work (and its characteristic features) for mining the steeply inclined (dip > 1 in 3) seams is more or less same with that of mining the inclined seams except that in case of mining steeply inclined seams, the footwall side is also advanced and kept benched for its stability and safety.



4.1.1.8.2. Sub-Surface Vein Type Deposits

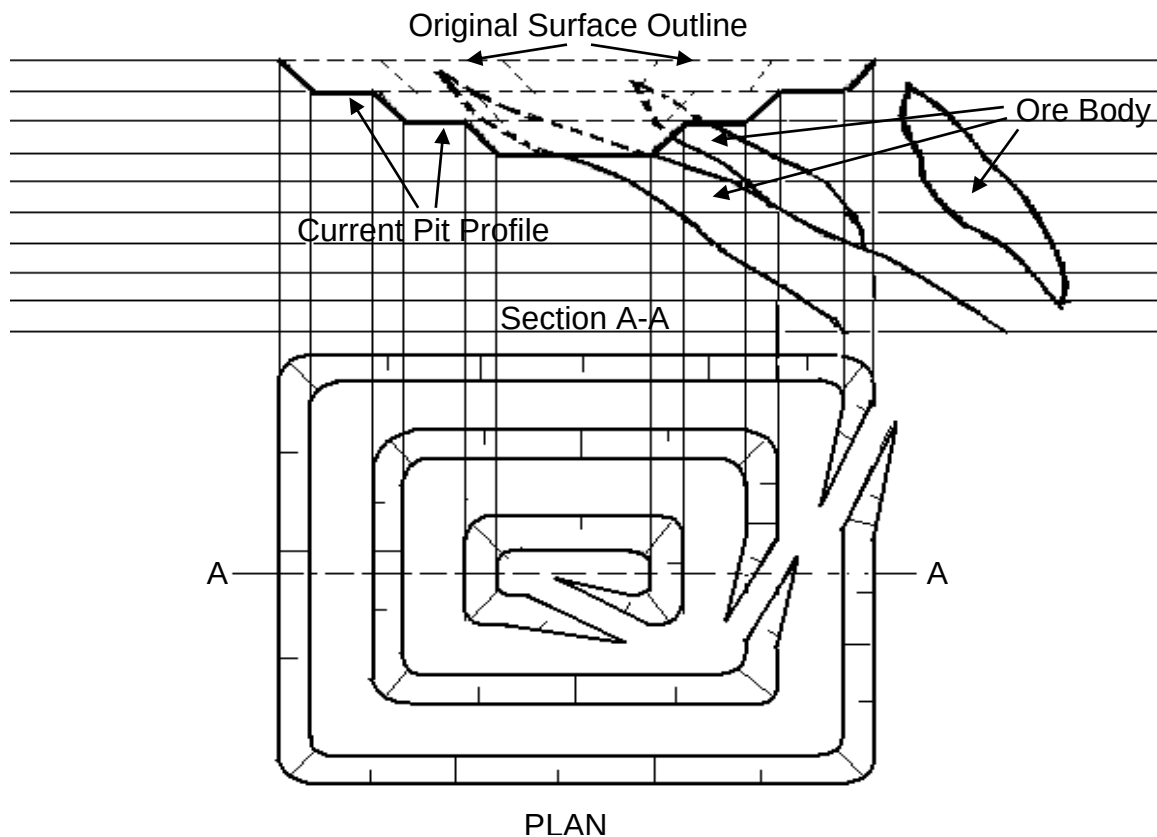
The method of work for mining of sub-surface vein type deposits (copper, lead & zinc, gold, etc.) differ considerably from the method of work for sub-surface bedded deposits.

A particular deposit may contain one or more number of veins. In both the cases, the box-cut is established in such a location that minimum excavation of overburden / waste rock is required to reach the ore body. In locating the box-cut care should be taken that it should not hinder the future expansion of the pit and should match with the overall pit plan. After establishing the box-cut the development of subsequent benches starts.

In this method the pit primarily advances towards the depth (vertical advance), and the advance in the sidewalls, footwall and the hang wall side in form of benches is made for stability of pit walls and for safety. The benches are generally started from the middle of the strike extent on the rise side of the vein(s) and have horizontal floor. With the progress of work, more and more benches are opened up extending the pit vertically downwards along with the expansion of pit plan in

outward direction from the centre. Opening up of benches continues till the final pit depth is reached. The ore is transported to ore crusher and the waste rock is transported to outside spoil dump through the established haul road ramp system.

The pit layout and extents depends on the number of veins, and their thickness, dip, continuity in dip direction, continuity in strike direction and respective spatial location. Depending on the number of veins and their respective features, there may be number of sub-pits.



4.1.1.7.3 Sub-Surface Pipe Type Deposits

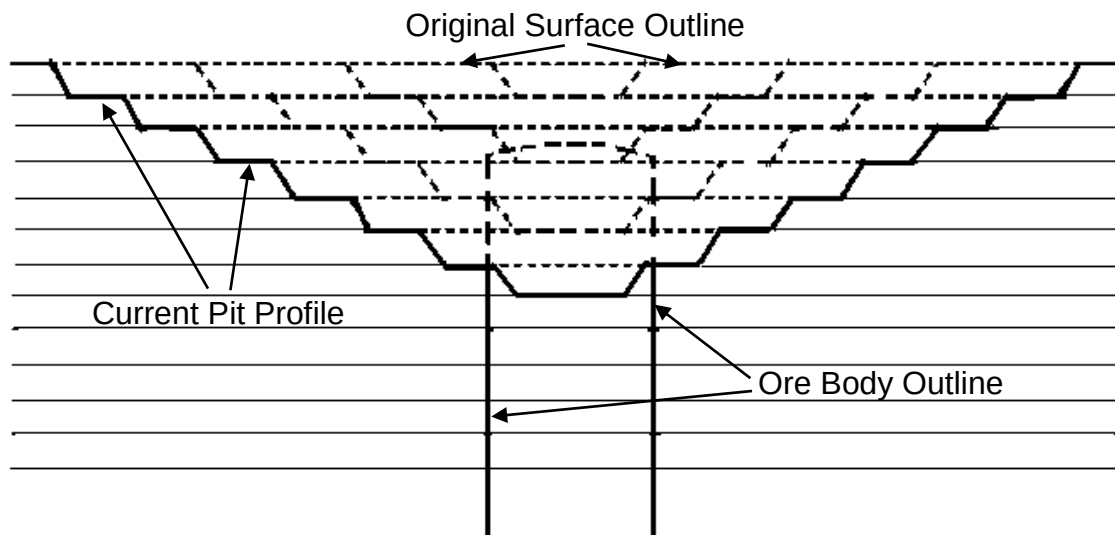
The method of work for mining of sub-surface pipe type deposits (such as, diamond bearing kimberlite or lamproite pipes) is somewhat similar to that of sub-surface vein type deposits.

The box-cut is located in such a manner that the top of the pipe shaped ore body at its central portion of the plan/area extent is reached as quickly as possible with possible minimum excavation of overburden/ waste rock, and its location matches with the overall pit plan. The development of subsequent benches starts after completion of the box-cut.

In this method also the pit primarily advances towards the depth (vertical advance), and the lateral advance of the pit walls in form of benches is made for stability and safety. The benches are generally started from the middle of the ore bearing zone that move outward and have horizontal floor. With the progress of work, more and more benches are opened up extending the pit vertically downwards along with the expansion of pit plan in outward direction from the centre.

Opening up of benches continues till the final pit depth is reached. The ore is transported to ore crusher and the waste rock is transported to outside spoil dump through the established haul road ramp system.

The pit layout is generally somewhat circular in case of vertical pipes and elliptical in case of non-vertical pipes. However, for a given deposit the pit layout and extent depends on number of pipes being mined, and their effective diameter, dip, continuity in dip direction and respective spatial location.

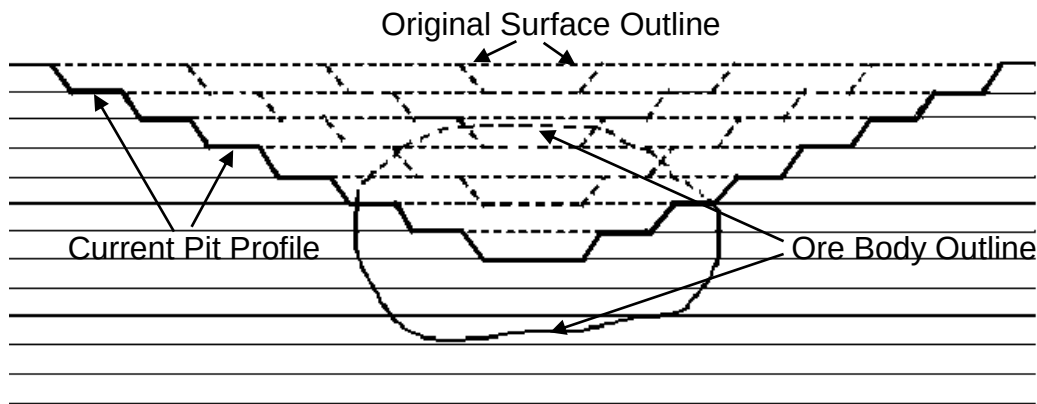


4.1.1.7.4 Sub-Surface Massive Deposits

The method of work for mining of sub-surface massive deposits is much similar to that of sub-surface pipe type deposits.

In this method also the pit primarily advances towards the depth (vertical advance), and the lateral advance of the pit walls in form of benches is made for stability and safety. The benches are generally started from the middle of the ore bearing zone that move outward and have horizontal floor. With the progress of work, more and more benches are opened up extending the pit vertically downwards along with the expansion of pit plan in outward direction from the centre. Opening up of benches continues till the final pit depth is reached. The ore is transported to ore crusher and the waste rock is transported to outside spoil dump through the established haul road ramp system.

The pit layout and the extent depend on the number of ore bodies, and their shape, size and respective spatial location.



4.1.1.7.5 Hilly Massive Deposits

The method of work for hilly massive deposits (such as, hilly iron ore deposits of India) differs in some aspects from the methods of work adopted for sub-surface deposits.

In mining of a hilly massive deposit, at first a road is developed upto the top of the hill maintaining the required gradient. This is followed by establishing the box-cut at the topmost bench floor level contour for development of that bench. The bench face is then moved forward extracting both ore and waste present in that bench and creating a horizontal surface at its floor level. Once the topmost bench has been moved sufficiently, the next lower bench is opened up from the contour of that bench floor level and the bench face starts moving forward. As the bench face progresses, it removes a slice (equal to bench height/thickness) of the hill by extraction of ore and waste. Finally when the whole slice is removed the bench ceases to exist. As more and more lower benches are opened up the top benches are completed in a top down order, and thereby gradually reducing the height of the hill top surface.

The same process is followed for opening and working of successive lower benches till the surface of the existing topmost bench levels with the surrounding surface at the foothill. Thereafter

the method of further pit development and working is similar to that of sub-surface massive deposits, which continues till the ultimate pit configuration is reached.

The ore is transported to ore crusher and the waste rock is transported to outside spoil dump through the established haul road ramp system. In the initial years of the mine life the material flow is in downward direction as the crusher, ore treatment facilities and waste dumps are generally located on the available flat ground at the foothill. The distance of the downward transport gradually decreases with the completion of successive top benches. Finally when the pit extends below the surrounding level the material flow direction is reversed, and the distance of transportation continue to increase thereafter till the end of pit life.

